

Software Engineering and Testing. BSC Year 2, 2024/2025

Assessment 4: Final Tested Product / Test Plan / Test Suite / Demo

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Declaration

I hereby certify that this material, which I now submit for assessment on the programme of study leading to the award of Ordinary Degree in Computing in the Institute of Technology Blanchardstown, is entirely my own work except where otherwise stated.

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Title: Pawsible

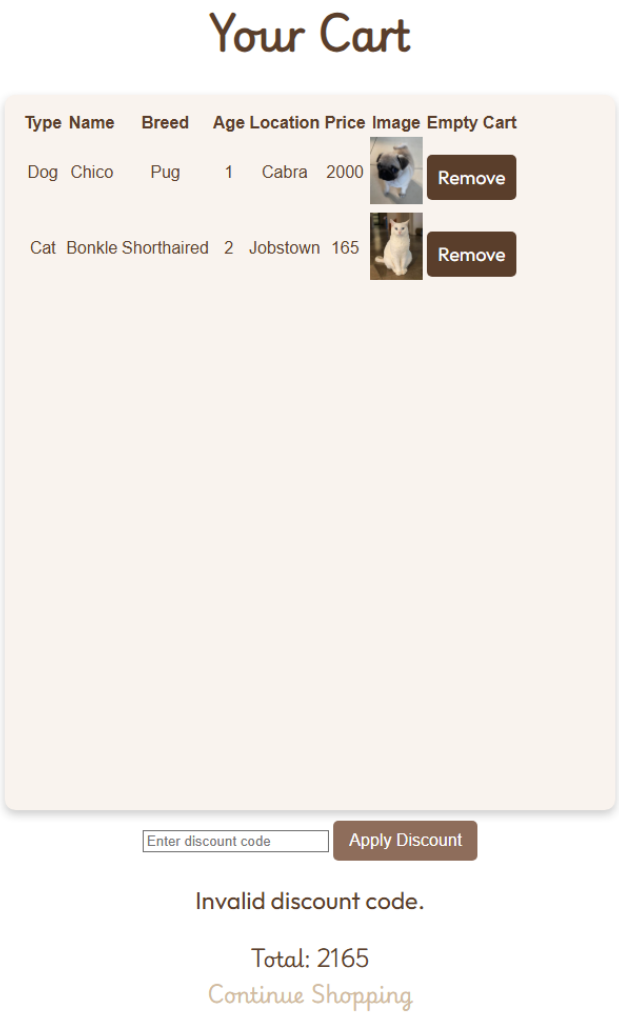
(A) User Interface Testing

User Guidance

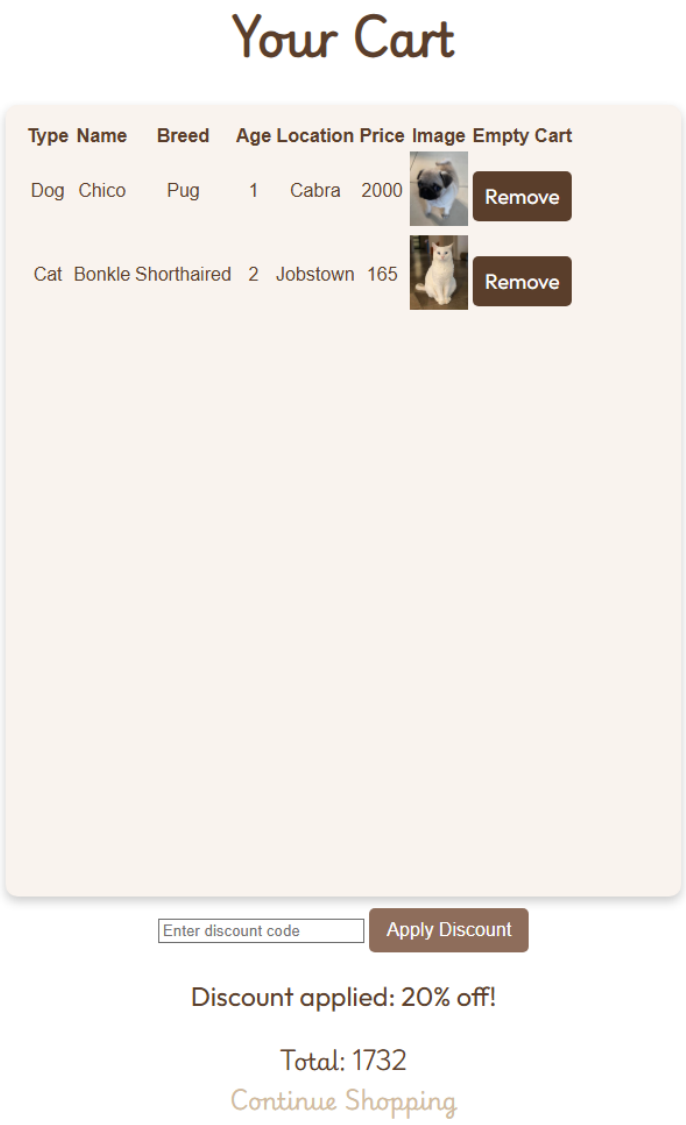
Feedback when error occurs – adoption application and discount code

On the cart page, in the event of an error such as an invalid voucher code being submitted, text will pop up below the discount text field informing the user that their code was invalid.

After entering voucher code “randomcode”



After inputting PAWSIBLE20



Consistency

Across the Pawsible website we use the same css files on all pages to keep a consistent look, with other pages having their own css files for elements specific to that page. To demonstrate, below are screenshots of the Pawsible home, browse and pet care guide pages. Please observe the colour scheme and font choices stay consistent across the website, along with php templates for the header and footer to keep them consistent across pages with DRY code.

Home

Pawsible

About View Available Pets Pet Care Guide Sponsor Contact

Welcome, you are logged in as: sean

Log out

Take a look at our most adopted pets!

- Corgi
- Pug
- Labrador
- Husky
- Shih Tzu







Want to see more? Browse our pets available for adoption near you here!

Interested in adopting one of our furry friends? Fill out our adoption application!

About Pawsible

Pawsible is dedicated to finding loving homes for cats and dogs in need. Our mission is to ensure that every pet has a chance to find their furever home.

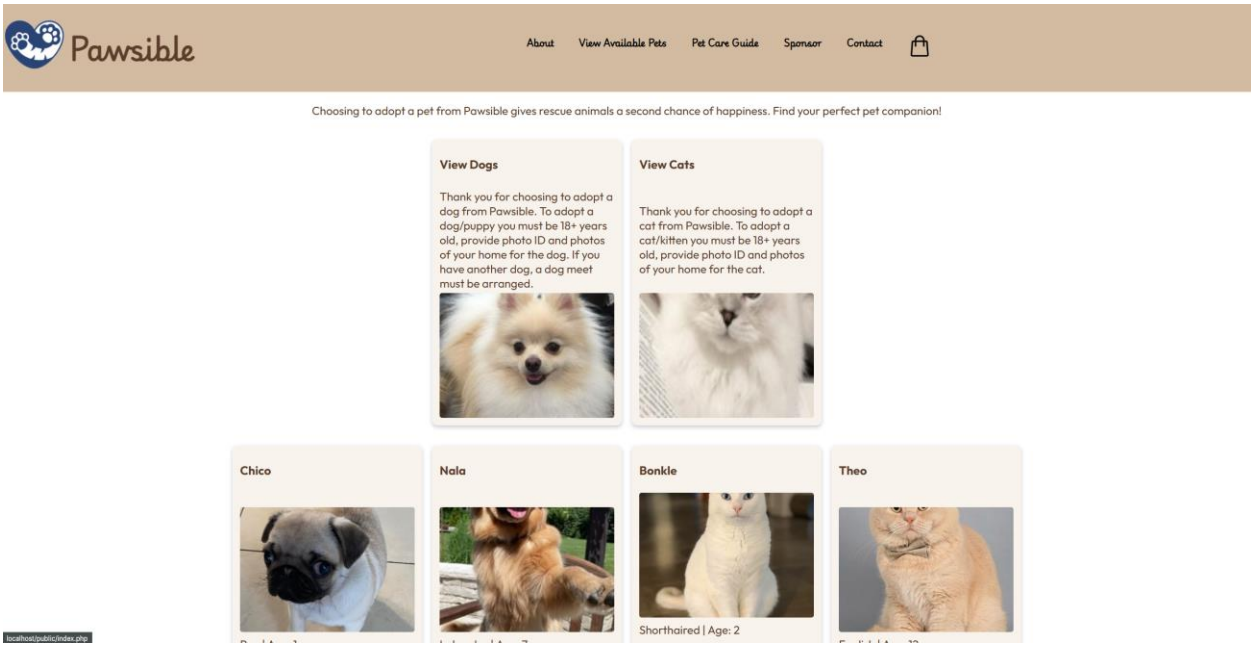
We work remotely with a network of shelters and

How We Facilitate Rehoming

At Pawsible, we follow a thorough process to ensure that each pet is matched with the right home:

- Initial Assessment: Through a thorough interview process, we assess each pet's health, behavior, and needs.

Browse



Pet Care Guide



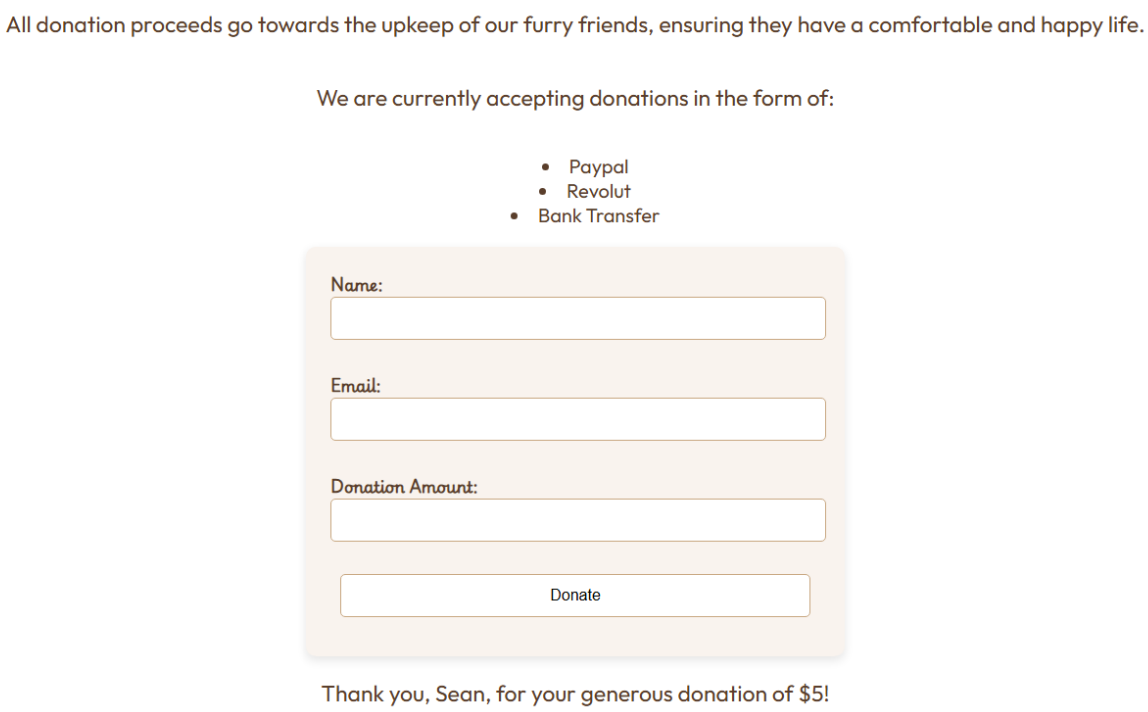
Feedback

Upon successfully filling out an application form, and donating; the system will provide the user with feedback confirming their form submission, printing the values the user inputted so they can review their submission and be assured the submission was successful.

Adoption application page



Donations page



Recoverability

In the event that a user adds a pet to their cart that they did not intend to (or they just changed their mind), there is a remove button next to the cart items to undo the mistake.

Minimal Clicks

The UI is designed to be as simple as possible, so users can easily navigate the site and perform their desired actions. For example, if a user intended to purchase Chico the dog, it would only take two clicks until they reach the checkout page from the home page. Home -> View Available Pets -> Add to cart (clicking this redirects the user to cart).

(B) Requirements Testing

Use Case 1: User Registration

Requirements Checklist:

	Completed
• User enters personal details (name, email, password)	✅
• System validates input (checks email format, password strength)	✅
• If valid, system creates an account and stores data	✅
• If invalid or already registered, system shows error or recovery options	❌

Use Case 2: Submit Request

Requirements Checklist:

	Completed
• User fills out personal details and message	✅
• System validates input (checks if fields are not empty, email valid)	✅
• System submits the request	✅
• Confirmation message displayed after submitting	❌

Use Case 3: Submit Adoption Application

Requirements Checklist:

	Completed
• User selects a pet and clicks “Adopt”	✅

- User fills out the adoption form ✅
- System validates input and stores the application ✅
- Admin reviews the application ❌

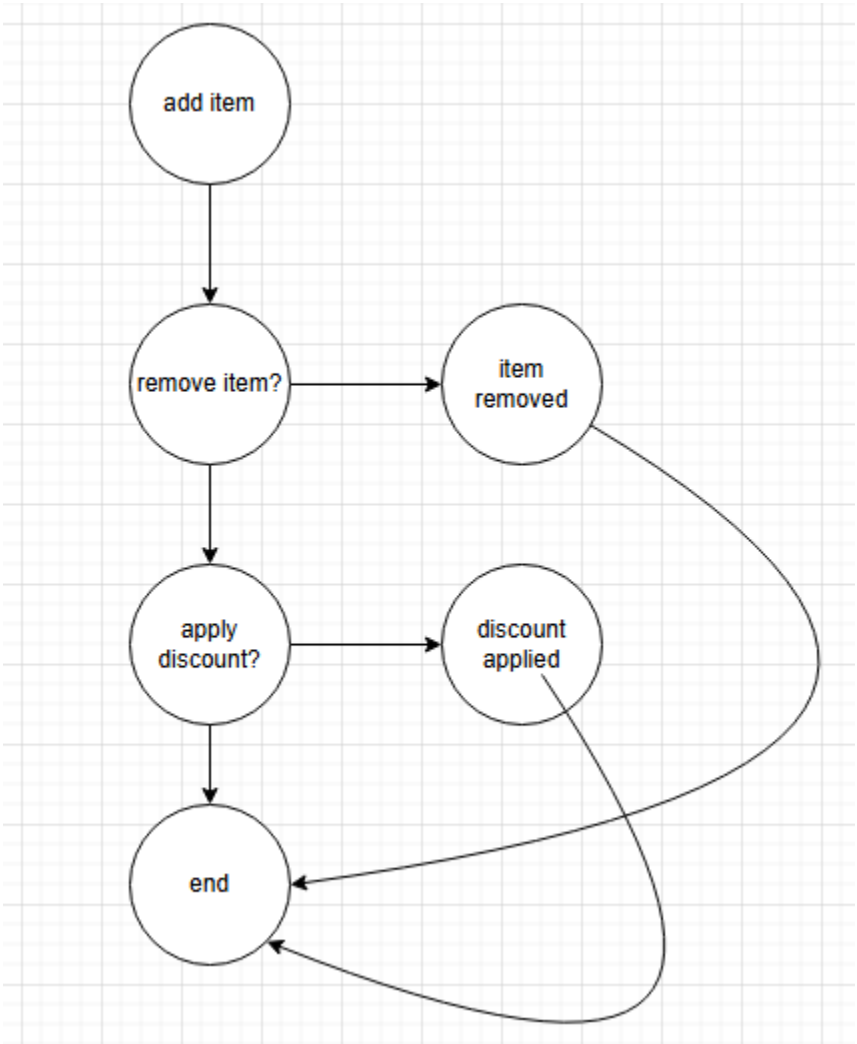
(C) Equivalence Partition Testing

Equivalence Partition	Test Input	Result
Valid username, email and password	username, email@gmail.com, password	User is sent to the login page to
Invalid email format	email	Please include @ in email
Empty email	Username, '', password	Please fill out this field
Empty username	'', email, password	Please fill out this field
Empty password	Username, email, ''	Please fill out this field

(D) Basis Path Testing

Test Cases	Steps	Result
TC1	Add pet to cart, end	Shows pet in cart at full price
TC2	Add pet to cart, remove item from cart, end	Cart will contain added pet and then become empty
TC3	Add pet to cart, apply voucher, end	Shows pet in cart with discounted price

Flow Chart Diagram



Cyclomatic Complexity Calculation

Edges – Nodes + 2

7 – 6 + 2 = 3